20240603 EB_MN WT-11, 5319, 4517 (DPR, qPCR)

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Status: In progress
Entry Created On: 27 Jun 2024 21:12:32 UTC
Entry Last Modified: 07 Jul 2026 18:45:30 UTC
Export Generated On: 01 Sep 2026 18:26:18 UTC

2024年6月3日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

Table1

	A	B	C
1	genotype	passage	wells
2	WT-11	P14+18	5
3	5319		5
4	4157		5

2024年6月5日 星期三

T2 -> T3

2024年6月7日 星期五

T3 -> T4 (~14 mL for over-weekend)

2024年6月10日 星期一

T4 -> T5

2024年6月12日 星期三

T5 -> T6

2024年6月14日 星期五

T6 -> T6 (~14mL for over-weekend)

2024年6月17日 星期一

T6 -> T7

Combine mine and Anouk's plates together

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table3

	A	B	C	D	E	F	G	H
1	cell	passage	trypsin	HIFBS/CTWM	cell count (M/mL)	volume	total	per-well (K)
2	WT11 (Anouk)	P32						
3	WT11 (YST)	P34	5 +0.5	11	3	6	18	
4	iPSC5319	PX+10	3	6	1.4	3	4.2	
5	iPSC4517	P12+9	3	6	0.215	3	0.645	215

WT-11

12 wells, 20uL virus/6 wells

PLATE A glass bottom for me (450K/well)

	1	2	3	4
A	110		111	
B				
C				

<u>For Anouk</u>

12 wells glass bottom plate (375K/well) + 6 wells 6-well plastic bottom plate (1.5M/well)

glass bottom plate for Anouk (375K/well)

	1	2	3	4
A	vLH92			
B				
C				

PLATE B plastic plate (1x = 200K)

	1	2	3	4
A	5319 2x	5319 1x	4517 1x	
B	5319 2x	5319 1x	4517 1x	
C	5319 2x	5319 1x	4517 1x	

2024年6月18日 星期二

Change T7 -> T7 if added virus

2024年6月19日 星期三

T7-> BP T8 if not added virus

2024年6月20日 星期四

T7 -> BP T8 if added virus

2024年6月21日 星期五

400uL BP T8 -> 500 uL BP T8

2024年6月24日 星期一

400uL BP T8 -> 500 uL BP T8

2024年6月26日 星期三

400uL BP T8 -> 500 uL BP T8

2024年6月27日 星期四

treat hypo 24h for PLATE B

2024年6月28日 星期五

PLATE B (RNA)

1. treat hypo 1 h for PLATE B
2. Remove media with suction
3. wash cells with cold PBS and remove the PBS
4. put the plate into LN
5. remove it from LN and place it on dry ice
6. store the plate in -80C

PLATE A -- imaging (DPR)

Treat SirDNA: (1:6000): remove 400uL and add 200uL BPT8 with SirDNA (600uL in total per well)

Drug prep: 100uL BP T8 with drug

Table2

	A	B	C	D	E	F	G	H
1		stock (mM)	working (uM)	dilution	total media	drug added	total drug added	added media volume
2	PR30	10	10	1000	1600	1.6	1.76	225
3			1	10000	1600	0.16		take 20 uL from above and add 185 uL T8
4	GR30	10	10	1000	1600	1.6	1.76	225
5			1	10000	1600	0.16		take 20 uL from above and add 185 uL T8
6	GP30	1	1	1000	1600	1.6	1.6	200

Run every thing all together with ND sequencing acquisition

Setting: New spinning disk confocal, 600x900, 16bit, HDR, 15 min in total

O.C.

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	640	16	10	100 ms
3	561	16	20	200 ms
4	488	16	50	1 s

plate map

	1	2	3	4
A	1	2	1	2
B	4	3	4	3
C	5	6	5	6

font color: 110, 111
acquisition group: group1, group2, group3
file name (cat) :

- pre-treatment: NT
- post-treatment: DPR

ND sequencing setting: **run every 8h, 4 counts in total**

File ID map

	A	B	C	D	E
1		0-1h	8-9h	16-17h	24-25h
2	group1	1-8	25-32	49-56	73-80
3	group2	9-16	33-40	57-64	81-88
4	group3	17-24	41-48	65-72	89-96

position ID

	A	B	C	D
1	condition #		110	111
2	1	NT	1,2	1,2
3	2	PR30 1uM	3,4	3,4
4	3	GR30 1uM	5,6	5,6
5	4	GP30 1uM	7,8	7,8
6	5	PR30 10uM	1,2	5,6
7	6	GR30 10uM	3,4	7,8

2024年7月17日 星期三

RNA extraction with PureLink™ RNA Mini Kit (quick protocol)

Lysis and Homogenization

1. Prepare lysis buffer: 3000uL lysis buffer + 30 uL 2-mercaptoethanol (100x)
2. Take the plate out and plate it on ice
3. Add 300uL lysis buffer for each well and plate the whole ice bucket on the shaker for 15min
4. Collect the lysis buffer into 1.5mL tubes
5. Combine column 1 and column2

PLATE B plastic plate (1x = 200K)1

	1	2	3	4
A	5319 2x	5319 1x	4517 1x	
B	5319 2x	5319 1x	4517 1x	
C	5319 2x	5319 1x	4517 1x	

6. Homogenization: Pass the lysate 5–**10** times through an **18**- to 21-gauge syringe needle.

RNA Purification

1. Add one volume 70% ethanol to each volume of cell homogenate.
   - 600 uL for 5319
   - 300 uL for 4517
2. Vortex to mix thoroughly and to disperse any visible precipitate that may form after adding ethanol.
3. Transfer up to 700 µL of the sample (including any remaining precipitate) to the spin cartridge (with the collection tube).
4. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through, and reinsert the spin cartridge into the same collection tube.
5. Repeat Steps 3–4 until the entire sample has been processed.
6. Add 350 µL Wash Buffer I to the spin cartridge.
7. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through.
8. Prepare DNaseI (10uL DNase I + 70 uL water, 80uL in total per rxn). Add 80 uL DNase I into the column and incubate for 15 min at RT.
9. Add 350 Wash Buffer I to the spin cartridge and centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through and the collection tube. Place the spin cartridge into a new collection tube.
10. Add 500 µL Wash Buffer II with ethanol to the spin cartridge.

11. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through.

12. Repeat Steps 8–9 once.

13. Centrifuge the spin cartridge at 12,000 × g for 1–**2** minutes to dry the membrane with bound RNA. Discard the collection tube and insert the spin cartridge into a recovery tube.

14. Add 30–100 µL (50uL) RNase–free water to the center of the spin cartridge.

15. Incubate at room temperature for 1 minute.

16. Centrifuge the spin cartridge for 2 minutes at ≥**12,000** × g at room temperature to elute the RNA from the membrane into the recovery tube. Note: If the expected RNA yield is >100 µg, perform 3 sequential elutions of 100 µL each. Collect the eluates in a single tube.

17. Store your purified RNA in -80 freezer.

RT with Thermo Scientific Maxima™ H Minus cDNA Synthesis Master Mix with dsDNase

1. Add the following reagents into a sterile, RNase free tube on ice in the following order:
(make total volumn into 12 uL)

image.png

10X dsDNase Buffer	1 µL
dsDNase	1 µL
Template RNA • Total RNA or • poly (A) mRNA or • Specific RNA	 1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng
Water, nuclease-free	to 10 µL
Total volume	10 µL

Table4

	A	B	C	D	E	F	G	H
1	**Sample**	**RNA conc (ng/ul)**	**A260/280**	**A260/230**	**RNA amount for 10 uL**	**minimum amount of RNA**	**volume for minimum RNA**	**Water up to 12 uL**
2	5319 hypo 1h	6	1.94	0.99	60	53	8.8	1.2
3	5319 hypo 24h	6.3	1.86	1.43	63	53	8.4	1.6
4	5319 2.5mM	7.3	2.03	1.28	73	53	7.3	2.7
5	4517 hypo 1h	5.3	1.81	0.5	53	53	10.0	0.0
6	4517 hypo 24h	9.8	1.82	0.35	98	53	5.4	4.6
7	4517 2.5mM	7.8	1.86	0.41	78	53	6.8	3.2

*note: the conc. may not be accurate. I measured the blank after measuring samples and got ~4 ng/uL.,,,

2. Mix gently and centrifuge.
3. Incubate at 37°C in PCR machine (or water bath) for 2 minutes.
4. Chill on ice, briefly centrifuge and place on ice.
5. prepare a master mix on ice: (modified the volume of water since we have 12 uL from the last step)

Table5

	A	B	C	D
1	reagent	volume (uL/rxn)	# of rxn	total volume
2	Maxima cDNA H Minus Synthesis Master Mix (5X)	4	6	26
3	Water, nuclease-free	4	6	26

6. Mix gently and centrifuge.
7. Set up PCR program:

Table6

	A	B
1	temp	time
2	25	10
3	50	15
4	85	5

8. The reaction product of the first strand cDNA synthesis can be used directly in qPCR. Store at –20°C for up to one week, or –70°C for long-term storage. Avoid freeze-thaw cycles of the cDNA.

2024年7月29日 星期一

QPCR

1. Dilute the cDNA 1:5 with nuclease-free water, to have 125 ul/sample total. You will use 3 ul of DNA/qPCR reaction.
2. Prepare the table to guide the loading of the qPCR.
3. Use this as template:
   The part with out highlight are Anouk's sameples

   5319 BP
   5319 1h Hypo
   5319 24h Hypo

   4517 BP
   4517 1h Hypo

4517 24h Hypo

Well1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-						
B	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP						
C	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-						
D	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	PFKP	cry_PFKP	ACTB												
E	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	STM2-	cry_STMN2													
F																								
G	ELAV3-	ELAV3-	ELAV3-	PFKP	PFKP	PFKP	ELAV3-	ELAV3-	ELAV3-															
H	ELAV3-	ELAV3-	ELAV3-	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-						
I	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP						
J	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-						
K	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB															
L	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-						
M	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP						
N	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-						
O	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	PFKP	cry_PFKP	ACTB												
P	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	STM2-	cry_STMN2													

Entrer the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

Table8

	A	B	C	D	E
1	**Reagent**	**Volume (ul) x1**	**Volume**	**Number of samples**	Number of genes to look at
2	*Water, PCR grade (vial 2, colorless cap)*	5	105	6	8
3	*PCR Primer mix 10x (5uM)*	2	42	**Total ul needed**	
4	*Master Mix, 2x conc. (vial 1, green cap)*	10	210	1680	
5	*Total volume*	17	357		
6			21		
7					
8					
9					

Load DNA first, then top up with 17ul of mix.

Run the default

Table7

	A	B	C
1	G1		24.86
2	G2		24.76
3	G3		24.65
4	G4		25.76
5	G5		25.66
6	G6		25.66
7	G7		25.52
8	G8		25.06
9	G9		25.08
10	H1		25.43
11	H2		25.48
12	H3		25.44
13	H4		26.28
14	H5		26.47
15	H6		26.3
16	H7		25.93
17	H8		25.94
18	H9		25.87
19	H10		35.47
20	H11		35.28
21	H12		34.83
22	H13		34.18
23	H14		35.7
24	H15		34.75
25	H16		35.53
26	H17		35.78
27	H18		34.63
28	I1		20.19
29	I2		20.31
30	I3		20.57
31	I4		21.1
32	I5		20.93
33	I6		20.71

34	I7		20.58
35	I8		20.66
36	I9		20.96
37	I10		33.48
38	I11		33.72
39	I12		33.87
40	I13		34.8
41	I14		33.23
42	I15		33.31
43	I16		34.82
44	I17		34.79
45	I18		35.87
46	J1		23.89
47	J2		23.72
48	J3		24.44
49	J4		24.65
50	J5		24.91
51	J6		24.73
52	J7		24.29
53	J8		24.53
54	J9		24.45
55	J10		29.75
56	J11		30.05
57	J12		29.99
58	J13		30.71
59	J14		31.27
60	J15		31.51
61	J16		28.94
62	J17		29.14
63	J18		29.13
64	K1		20.67
65	K2		20.58
66	K3		21.2
67	K4		21.31
68	K5		21.24
69	K6		21.51
70	K7		20.76

71	K8		20.76
72	K9		21.12
73	L1		25.36
74	L2		25.3
75	L3		25.52
76	L4		25.04
77	L5		25.32
78	L6		25.12
79	L7		27.31
80	L8		27.17
81	L9		27.25
82	L10		
83	L11		40
84	L12		38.7
85	L13		
86	L14		
87	L15		36.79
88	L16		38.25
89	L17		40
90	L18		40
91	M1		25.87
92	M2		25.86
93	M3		25.81
94	M4		25.69
95	M5		25.6
96	M6		25.51
97	M7		27.12
98	M8		27.22
99	M9		27.51
100	M10		37.24
101	M11		35.97
102	M12		35.57
103	M13		36.87
104	M14		35.58
105	M15		36.46
106	M16		36.66
107	M17		40

108	M18		40
109	N1		22.57
110	N2		22.49
111	N3		22.48
112	N4		22.09
113	N5		22.14
114	N6		22.07
115	N7		23.94
116	N8		24.1
117	N9		23.88
118	N10		35.54
119	N11		34.78
120	N12		34.92
121	N13		35.15
122	N14		34.73
123	N15		36.66
124	N16		35.78
125	N17		35.31
126	N18		36.72
127	O1		24.84
128	O2		24.75
129	O3		24.52
130	O4		24.02
131	O5		24
132	O6		24.47
133	O7		25.13
134	O8		25.82
135	O9		25.9
136	P1		21.83
137	P2		21.77
138	P3		21.52
139	P4		21.34
140	P5		21.49
141	P6		21.06
142	P7		22.87
143	P8		22.54
144	P9		23.02

145	P10		40

RT qPCR

Introduction

Freezing of cells, RNA extraction, RT and qPCR protocol with hMNs

Materials

- hMNs in culture, day 10 after plating
- ice cold PBS
- RNA mini Kit Invitrogen
- Needles 18G
- 1 mL syringe
- Maxima™ H Minus cDNA Synthesis Master Mix Catalog number: M1681
- Integra Biosciences Corp 3423, 125UL 10ECO RKS, 384TIPS
- Roche Diagnostics LIGHTCYCLER 480 SYBR GREEN I M
- Sapphire Microplate 384 well for qPCR

Procedure

Freezing of cells in plate

1. Once the treatment is done, remove medium from the cells
2. Optional: ice cold PBS wash (hMNs are very frigile, I usually avoid washing to make sure not to lose them)
3. Freeze the plate in liquid nitrogen and transfer to -80 until extraction.

RNA extraction

4. Follow RNA mini Kit Invitrogen user manual
5. Only change: added a DNAseI treatment! Do a first wash with Wash Buffer I with 350 ul. Then incubate 15' at RT with 10ul DNAseI in 80 ul water per sample. Then spin and another 350ul Wash Buffer I.
6. Elute in 50 ul and quantify with Nanodrop.

RT

7. Only maximum 10 ul can be used for the following step, so the RNA concentration is what limits your choice for RNA amount to start with. I have used even 50 ng total to start with success.

8. DNAse treatment in PCR tube:

Table1

	A	B	C
1	10X dsDNase Buffer	1 µL	
2	dsDNase	1 µL	
3	Template RNA •Total RNA or •poly(A) mRNA or •Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng	
4	Water, nuclease-free	to 12 µL	
5	Total volume	12 µL	

9. Prepare mix of

   1ul DNAse/sample +1

   1ul of buffer 10x/sample +1

10. Add 2 ul/ tube.

11. Mix gently and centrifuge.

12. Incubate at 37°C in a preheated thermomixer or water bath for 2 minutes.

13. Chill on ice, briefly centrifuge and place on ice.

14. Add the following components to the tube on ice:

Table2

	A	B	C
1	Maxima cDNA H Minus Synthesis Master Mix (5X)	4 µL	
2	Water, nuclease-free	4 µL	

15. Prepare mix of

    4ul Maxima cDNA H Minus Synthesis Master Mix (5X)/sample +1

    1ul of Water, nuclease-free/sample +1

16. Add 8 ul per tube.

17. Mix gently and centrifuge. Load the program in the PCR machine:

18. Incubate at 25°C for 10 minutes.

19. Incubate at 50°C for 15 minutes.

20. Terminate the reaction by heating at 85°C for 5 minutes.

21. Use directly or freeze in -80 for long term storage.

qPCR

22. Dilute the cDNA 1:5 with nuclease-free water, to have 125 ul/sample total. <u>You will use 3 ul of DNA/qPCR reaction</u>.

23. Prepare the table to guide the loading of the qPCR.

24. Use this as template:

    wt-11 BP

    wt-11 1h Hypo

    wt-11 24h Hypo

Table3

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-						
3	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP						
4	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-	cry_STMN2-						
5	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	PFKP	cry_PFKP	ACTB												
6	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	STM2-	cry_STMN2													
7																								
8																								
9																								
10																								
11																								
12																								
13																								
14																								
15																								
16																								
17																								

25. Entrer the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

Table4

	A	B	C	D	E
1	**Reagent**	**Volume (ul) x1**	**Volume x27**	**Number of samples**	Number of genes to look at
2	*Water, PCR grade (vial 2, colorless cap)*	5	30	3	8
3	*PCR Primer mix 10x (5uM)*	2	12	**Total ul needed**	
4	*Master Mix, 2x conc. (vial 1, green cap)*	10	60	480	
5	*Total volume*	17	102		
6					
7					
8					
9					

26. Load DNA first, then top up with 17ul of mix.

27. Run the default

28.